

Homeplate v1.0 — Build Plan

Derived from **SRS v3.2** (frozen baseline), **SPMP v1.0**, and **ADR-001...006**. Companion artifact: `requirements-inventory.json` (all 14 FR + 13 NFR + 8 AB with executable acceptance criteria).

Repository state at planning time: **green-field for application code**. The repo contains only **ADRs/**, **Weekly Report/**, `.claude/workflows/homeplate-build.js` and `docs/_generated/`. No `package.json`, `src/`, `client/`, migrations or tests exist yet. Everything below is a new build.

1. Stack

Concern	Choice	Why
Client	React 18 + Vite, responsive single-page web app	SRS §2.1.2 (SPMP §5.2.1's "React Native" loses to the SRS)
API	Node.js 20+ / Express 4, stateless REST over HTTPS/JSON	SRS §2.4 fixed constraint
Language	JavaScript (CommonJS on the server, ESM in the client)	No transpile step; Jest + Supertest work out of the box (SRS §4.1)
Source of truth	PostgreSQL 16 via <code>pg</code> connection pool, parameterized SQL only	SRS §2.4, NFR-11
Cache / sessions	Redis 7 via <code>ioredis</code> — sessions and read cache only	SRS §2.4, ADR-001, ADR-006
Media	S3-compatible object storage (MinIO locally) via <code>@aws-sdk/client-s3</code> , referenced by key	ADR-004
Deferred work	PostgreSQL outbox table + in-process polling worker, per-service adapters	ADR-001 / ADR-003
Validation	<code>zod</code> schemas at the API boundary + shared sanitizer	NFR-11, ADR-006
Passwords	Argon2id via <code>@node-rs/argon2</code> (prebuilt binaries). Documented fallback: <code>bcryptjs</code> cost 12 if the prebuilt binary is unavailable on the build host — record it as a deviation	NFR-04
Sessions	Opaque 256-bit token in an <code>HttpOnly/Secure/SameSite=Lax</code> cookie; session record in Redis with TTL	ADR-006
Moderation LLM	HTTPS adapter, provider configurable by env; default Anthropic Messages API (<code>MODERATION_MODEL=claude-opus-5</code>) with a JSON-schema structured output returning <code>{category, confidence}</code>	ADR-002, FR-08

Concern	Choice	Why
Maps	Google Maps/Places Geocoding + Places over HTTPS behind one adapter, results cached in Redis	ADR-005
Email / push	SendGrid (@sendgrid/mail) and Firebase Cloud Messaging (firebase-admin), worker-only	SRS §2.1.3, FR-13, FR-07
Server tests	Jest + Supertest against a seeded test database; k6 (load), axe-core (a11y), OWASP ZAP baseline (ST-04)	SRS §4
Client tests	Vitest + @testing-library/react + axe-core	NFR-07
Package manager	npm (Node 20+); client installed with npm --prefix client	SPMP §5.1.3 free tier
Local infra	Docker Compose: PostgreSQL, Redis, MinIO	SPMP §5.1.3, ADR-004

Directory layout

db/migrations/	numbered .sql migrations (source of truth for the §3.4 schema)
scripts/	migrate.js, seed.js, worker.js, retention-sweep.js, gen-dev-certs.sh
src/config/	env schema, fail-fast loader, locale/AB 626 caps (SRS §2.1.7)
src/db/	pg pool, withTransaction, redis client
src/lib/	logger, errors, resilience (timeout/retry/backoff), httpClient, cache, sanitization
src/middleware/	requestContext, errorHandler, security (TLS/HSTS), validate
src/routes/	router registry that mounts src/modules/*/routes.js
src/adapters/	maps, sendgrid, fcm, llmModeration, objectStorage (worker-side only)
src/outbox/	outbox.enqueue, dispatch registry, worker; handlers/*.js
src/modules/	auth, users, eligibility, media, listings, search, hosts, bookings, reviews, messaging, moderation, safety, privacy, notifications
src/app.js, src/server.js	Express app factory and HTTPS bootstrap
client/	React + Vite responsive web app (own package.json)
tests/unit/	implementer-written unit/integration tests
tests/<lane>/	verifier lanes (tc-core, tc-booking, it-adapters, st-security, rt-lt-resilience, mt-ut-quality, adr-conformance, coverage)
tests/fixtures/	seed + the versioned >=200-item moderation evaluation set (IT-03)
docs/_generated/	this plan, the requirement inventory, extracted SRS/SPMP text

Two conventions that make parallel work safe

1. **Route registry (src/routes/index.js, owned by U1-HTTP).** It resolves the known list of src/modules/<name>/routes.js paths, mounts each one that exists, and logs a startup warning for any that does not. Real logic, no stub — later waves drop their router in and it is mounted with no edit to a shared file.
2. **Outbox handler discovery (src/outbox/dispatch.js, owned by U1-OUTBOX).** The dispatcher reads src/outbox/handlers/*.js at startup; each file exports { type, handle(payload, ctx) }. Feature units own individual handler files, so no shared registry file is ever edited twice.
3. **Client route discovery (client/src/App.jsx, owned by U5-SHELL).** Routes are collected with Vite's import.meta.glob('./features/*/routes.jsx', { eager: true }), which is valid with zero matches. Wave-6 feature units add their own routes.jsx and are auto-mounted.

2. Commands

Purpose	Command
install	<code>npm install && npm --prefix client install</code>
infra	<code>docker compose up -d</code> (PostgreSQL 5432, Redis 6379, MinIO 9000/9001)
migrate	<code>npm run migrate</code> (applies <code>db/migrations/*.sql</code> in order, records them in <code>schema_migrations</code>)
seed	<code>npm run seed</code> / <code>npm run seed:volume</code> (NFR-02 scale for LT-02)
build	<code>npm run build</code> (client production bundle)
test	<code>npm test</code> (Jest: <code>src/** + tests/**</code> ; client suite: <code>npm run test:client</code>)
lint	<code>npm run lint</code> (ESLint + Prettier over server and client)
dev	<code>docker compose up -d && npm run dev</code> (API + outbox worker + Vite client concurrently)
load	<code>npm run test:load</code> (k6, LT-01/LT-02)

TLS for local development: `scripts/gen-dev-certs.sh` writes a self-signed cert into `certs/` (git-ignored). The server binds HTTPS with `minVersion: 'TLSv1.2'` and refuses plain HTTP with 403 `HTTPS required`; Supertest exercises the Express app directly with transport enforcement switched off by an explicit config flag that fails closed when `NODE_ENV=production`.

3. Waves

A wave is a dependency layer: wave N may assume waves $1 \dots N-1$ exist on disk. Within a wave, units are file-disjoint and independently buildable — no unit's tests import a sibling's files.

Wave 1 — Foundation (6 units)

Unit	Title	Requirements	Owens
U1-CONFIG	Config + locale/AB 626 caps	NFR-13, FR-11, FR-12, NFR-12	<code>src/config/{index,schema,locale}.js</code> , <code>.env.example</code> , <code>tests/unit/config.test.js</code>
U1-DB	Schema, migrations, DB/Redis clients, cache helper	NFR-02, NFR-13, NFR-11, NFR-01	<code>db/migrations/0001_core_schema.sql</code> , <code>db/migrations/0002_indexes_constraints.sql</code> , <code>scripts/{migrate,seed}.js</code> , <code>src/db/{pool,tx,redis}.js</code> , <code>src/lib/cache.js</code> , <code>tests/unit/db.test.js</code>

Unit	Title	Requirements	Owens
U1-OBS	Logging, errors, correlation IDs, resilience + HTTP client	NFR-08, NFR-09	src/lib/{logger,errors,resilience,l src/middleware/{requestContext,err tests/unit/observability.test.js
U1-VALID	Input validation and sanitization layer	NFR-11, AB-06	src/middleware/validate.js, src/lib/sanitize.js, src/schemas/common.js, tests/unit/validation.test.js
U1-HTTP	Express app, TLS/HTTPS enforcement, route registry	NFR-03, NFR-08, AB-05	src/app.js, src/server.js, src/middleware/security.js, src/routes/index.js, scripts/gen-dev-certs.sh, tests/unit/app.test.js
U1-OUTBOX	Transactional outbox + worker skeleton	FR-13, NFR-09, NFR-08	db/migrations/0003_outbox.sql, src/outbox/{outbox,dispatch,worker} scripts/worker.js, tests/unit/outbox.test.js

U1-DB owns the whole SRS §3.4 schema — users, host_profiles, email_verification_tokens, listings, bookings, reviews, messages, safety_alerts, moderation_decisions, moderation_queue, notification_attempts, media_objects, data_requests, access_log — plus the invariants downstream units depend on: unique users.email; unique (host_id, local_date) on non-cancelled listings (FR-11/AB-07); CHECK (seats_remaining >= 0 AND seats_remaining <= seat_capacity) (FR-12/FR-14); CHECK (rating BETWEEN 1 AND 5); moderation-status enums defaulting to pending for listings and reviews. U1-OUTBOX owns its own migration (outbox_jobs with type, payload jsonb, dedupe_key, available_at, attempts, status, last_error) so the two units never touch the same file.

Wave 2 — Platform services and external adapters (5 units)

Unit	Title	Requirements	Owens
U2-IDENTITY	Auth, sessions, login rate limiting, registration, email verification, profile	FR-10, NFR-04, NFR-05, NFR-06, AB-05, AB-07	src/modules/auth/{passwords,session src/modules/users/{repo,service,tol tests/unit/identity.test.js
U2-ELIGIBILITY	The single eligibility policy interface	FR-09, NFR-06	src/modules/eligibility/{policy,rep tests/unit/eligibility.test.js
U2-ADAPTERS-COMMS	SendGrid + FCM adapters (timeout/retry/fallback)	FR-13, FR-07, NFR-09	src/adapters/{sendgrid,fcml}.js, tests/unit/adapters-comms.test.js
U2-ADAPTER-MAPS	Google Maps/Places adapter + Redis geo/result cache	FR-01, NFR-01, NFR-09	src/adapters/maps.js, tests/unit/adapters-maps.test.js
U2-MEDIA-LLM	Object-storage adapter (per-object delete) + LLM moderation adapter + media service	FR-02, FR-03, FR-05, FR-08, NFR-10, NFR-12	src/adapters/{objectStorage,llmMod src/modules/media/{repo,service}.js tests/unit/adapters-media-llm.test

Registration and login live in **one** unit because registration needs the password hasher and the session issuer; splitting them would make one unit's tests depend on a concurrently-built sibling. Adapters are pure library

code with no Express routes — nothing here is reachable from a request handler except through the worker (ADR-001).

Wave 3 — Core marketplace (4 units)

Unit	Title	Requirements	Owns
U3-LISTINGS	Listing create/update/cancel with MEHKO caps; meal detail	FR-11, FR-02, AB-01, AB-03, AB-07	src/modules/listings/{repo,rules,se tests/unit/listings.test.js
U3-SEARCH	Discovery by location/time/host/cuisine with Redis cache and degraded mode	FR-01, NFR-01, NFR-09	src/modules/search/{repo,service,r tests/unit/search.test.js
U3-HOSTS-MEDIA	Host profile page + media upload/delete API	FR-03, NFR-13, AB-08	src/modules/hosts/{repo,service,rou src/modules/media/routes.js, tests/unit/hosts-media.test.js
U3-BOOKINGS	Atomic reservation, cancellation, completion lifecycle, notification enqueue	FR-12, FR-13, FR-14, FR-04, AB-02	src/modules/bookings/{repo,service, tests/unit/bookings.test.js

U3-BOOKINGS is the core loop (SPMP WA-3, never cut). It asserts the booking row and its outbox rows commit in one transaction, that a rejected reservation leaves capacity untouched, and that no adapter is called on the request path. `lifecycle.js` also owns the scheduled `pending → in progress` transition at the listing's start time, run as a periodic worker job.

Wave 4 — Trust, safety and data lifecycle (5 units)

Unit	Title	Requirements	Owns
U4-REVIEWS-MESSAGING	Mutual reviews + booking-scoped messaging	FR-05, FR-06, AB-04	src/modules/reviews/{repo,service,r src/modules/messaging/{repo,service tests/unit/reviews-messaging.test.js
U4-MODERATION	Two-stage pipeline, moderator queue, publication policy, IT-03 eval set	FR-08, NFR-10, AB-01, AB-03, AB-04	src/modules/moderation/{prefilter,p src/outbox/handlers/moderationScan tests/fixtures/moderation-eval/v1/* tests/unit/moderation.test.js
U4-SAFETY	Safety alerts with moderator notification and retried email delivery	FR-07	src/modules/safety/{repo,service,r src/outbox/handlers/safetyAlert.js, tests/unit/safety.test.js
U4-PRIVACY	Account deletion, media erasure by key, CCPA export, retention sweep	NFR-12, NFR-13, AB-08	src/modules/privacy/{repo,service, src/outbox/handlers/accountErasure scripts/retention-sweep.js, tests/unit/privacy.test.js
U4-NOTIFY	Worker dispatch paths for booking notifications and email verification	FR-13, NFR-08, NFR-09	src/outbox/handlers/{bookingNotifi src/modules/notifications/{repo,ser tests/unit/notifications.test.js

Producers (waves 2–3) assert that the outbox row is written; consumers (this wave) implement and test delivery against mocked adapters. Each handler owns its own file inside `src/outbox/handlers/`, so ownership stays exclusive while the dispatcher discovers them all.

Wave 5 — Client foundation (3 units)

Unit	Title	Requirements	Owns
U5-SHELL	Vite app, router with glob-based feature registry, layout, ally baseline, test harness	NFR-07, SRS §2.1.2	<code>client/package.json</code> , <code>client/vite.config.js</code> , <code>client/index.html</code> , <code>client/vitest.setup.js</code> , <code>client/src/{main.jsx,App.jsx,router}</code> , <code>client/src/styles/global.css</code> , <code>client/src/__tests__/app.test.js</code>
U5-API-AUTH	Typed API client, session/auth context, error + degraded-mode surfacing	NFR-03, NFR-09, AB-05	<code>client/src/lib/{api.js,errors.js}</code> , <code>client/src/auth/{AuthProvider.jsx,u}</code> , <code>client/src/components/DegradedBanne</code>
U5-UI-KIT	Accessible primitives (form field, button, rating, gallery, pagination, skeleton)	NFR-07	<code>client/src/lib/__tests__/api.test.js</code> , <code>client/src/components/ui/*.jsx</code> , <code>client/src/components/ui/__tests__/</code>

Wave 6 — Client features (4 units)

Unit	Title	Requirements	Owns
U6-DISCOVERY	Search/browse, meal detail, host profile	FR-01, FR-02, FR-03, NFR-07, NFR-09	<code>client/src/features/discovery/**</code>
U6-BOOKING	Reserve a seat, my bookings, cancel, dual completion confirmation	FR-12, FR-14, FR-04, FR-13, AB-02	<code>client/src/features/booking/**</code>
U6-COMMUNITY	Reviews, booking chat, safety-alert trigger	FR-05, FR-06, FR-07, AB-04	<code>client/src/features/community/**</code>
U6-ACCOUNT-MOD	Signup/verify/login, profile + host onboarding, data export/deletion, moderator queue	FR-10, FR-09, FR-08, NFR-06, NFR-12, NFR-13	<code>client/src/features/account/**</code> , <code>client/src/features/moderation/**</code>

Each feature directory contains its own `routes.jsx` exporting route objects, which the shell picks up automatically.

4. Binding invariants every unit is checked against

1. **No inline adapter calls.** Request handlers may not import `src/adapters/*`. Only `src/outbox/handlers/*` and worker code may. (ADR-001/003; verified by the `adr-conformance` lane.)

2. **One transaction, no dual writes.** A business row and its outbox row commit together via `withTransaction`; the outbox payload carries IDs only, never names, emails or phone numbers.
3. **Pending until approved.** Listings and reviews default to `moderation_status='pending'` and are filtered out of every public read path. A moderation outage leaves them pending forever — it must never publish unreviewed content. Messages deliver immediately and are scanned asynchronously.
4. **One eligibility interface.** `canReserveSeat` / `canPublishListing` exist once, in `src/modules/eligibility/policy`. No module re-implements them.
5. **Media by key.** Listing/review/profile media live in object storage, referenced from PostgreSQL by key, deleted per-object during account erasure.
6. **Redis holds sessions and cache only.** No business state whose loss would change a booking, listing or moderation outcome.
7. **HTTPS/TLS 1.2+ only**, validation at the API boundary, no MFA / ID verification / payments.

5. Mapping to the SPMP work activities

SPMP activity	Units
WA-1 Auth & eligibility	U2-IDENTITY, U2-ELIGIBILITY
WA-2 Discovery/listing + Maps	U3-LISTINGS, U3-SEARCH, U2-ADAPTER-MAPS
WA-3 Booking (outbox, atomic capacity)	U3-BOOKINGS, U1-OUTBOX
WA-4 Review & messaging	U4-REVIEWS-MESSAGING
WA-5 Safety alert	U4-SAFETY
WA-6 Data lifecycle	U4-PRIVACY
WA-7 Moderation integration	U4-MODERATION, U2-MEDIA-LLM (LLM adapter)
WA-8 Media storage adapter	U2-MEDIA-LLM, U3-HOSTS-MEDIA
WA-9 Web client	U5-, U6-
WA-10 Worker/outbox dispatcher + adapters	U1-OUTBOX, U2-ADAPTERS-COMMS, U4-NOTIFY
WA-11 Test suite vs SRS §4	verification lanes (<code>tests/<lane>/</code>)

6. Open questions for the team

Each is resolved here with the reading most faithful to the SRS, and flagged for confirmation.

1. **React Native vs React web.** SPMP §5.2.1/§6.2 name React Native; SRS §2.1.2 mandates a single responsive React *web* app and states no native app ships in v1.0. **SRS wins — web only.**
2. **AB 626 numeric caps are not in the SRS.** SRS §2.1.7 says the caps are configuration. Defaults used: 1 listing per host per day (stated in §3.4), 30 meals/day and 60 meals/week (the actual AB 626 limits). Team must confirm the numbers and the legal review note.
3. **Per-guest concurrent pending-booking limit (FR-12) has no stated value.** Default 3, configurable.
4. **Notification channel.** FR-13 says “the configured notification”. Web push via FCM needs a service worker and VAPID key the team does not have; **email (SendGrid) is the default channel and the FCM adapter ships behind a config flag**, satisfying “configured” without blocking the demo.

5. **Moderation LLM provider is not free-tier.** The adapter is provider-agnostic (`LLM_MODERATION_BASE_URL`, `LLM_MODERATION_API_KEY`, `MODERATION_MODEL`) so CI and the automated test suite run against a deterministic mock; IT-03's 200-item measurement needs one real key. Team must decide who supplies it.
6. **The ≥ 200 -item labelled evaluation set does not exist yet.** U4-MODERATION authors and versions it; it must be reviewed by a human before the NFR-10 result is claimed.
7. **Encryption at rest (NFR-13).** Free-tier PostgreSQL offers no TDE. Plan: application-level AES-256-GCM for phone and emergency-contact columns with a key from the environment, plus a documented volume-encryption assumption for the rest. Confirm this satisfies ST-06.
8. **Backup expiry (NFR-12).** No managed backup service exists in the academic environment. Plan: `scripts/retention-sweep.js` plus a documented 30-day retention policy; ST-05 verifies it as configuration review, not as a live backup deletion.
9. **pending $\rightarrow$ in progress transition (FR-04/§3.4)** is not attached to a user action. Implemented as a periodic worker job at the listing's scheduled start (the outbox mechanism is event-driven; this is a scheduled sweep on the same worker process).
10. **Emergency contact is optional (§3.4).** If absent, FR-07 still persists the alert and notifies the moderator; delivery status is recorded as `no_channel` rather than failed.
11. **Host address exposure.** The SRS requires location for discovery (FR-01) but also data minimization (NFR-13). Plan: approximate area pre-booking, exact address released only to a confirmed guest of that booking.
12. **NFR-01 at 200 concurrent users** cannot be honestly demonstrated on arbitrary laptop hardware. LT-01/LT-02 report measured numbers; a run that cannot reach 200 VUs is marked untestable with the achieved level, never reported as a pass.
13. **UT-01's 5-participant moderated usability study** is a human activity outside automation.